

1 Teleconnection attribution

1.1 Background

Large-scale climate modes influence lake thermal regimes through atmospheric and oceanic teleconnections. We evaluate how five major teleconnection indices—ENSO (Nino3.4), PDO, AMO, NAO, and AO—contribute to the observed STL trend and its first difference across the eight temporal response types.

The analysis proceeds in two levels: (1) regression of the annual STL trend on the five indices, and (2) regression of the first-differenced STL trend on the same indices. Level 1 captures which indices drive the long-term trend level; Level 2 captures which indices drive year-to-year changes in trend slope.

1.2 Level 1: Trend attribution

i Finding 14

AMO is the dominant driver of STL trend across all clusters. The global mean standardised coefficient is $\beta = 0.56$, with a partial R^2 of 0.34. PDO is the second strongest driver ($\beta = -0.18$, partial $R^2 = 0.05$). ENSO, NAO, and AO have weak independent effects.

i Finding 15

AMO dominance varies by cluster. C5 (warming Europe) has the highest AMO coefficient ($\beta = 0.69$, partial $R^2 = 0.49$). C4 (decelerating North America) has the lowest AMO effect but the strongest PDO effect ($\beta = -0.31$).

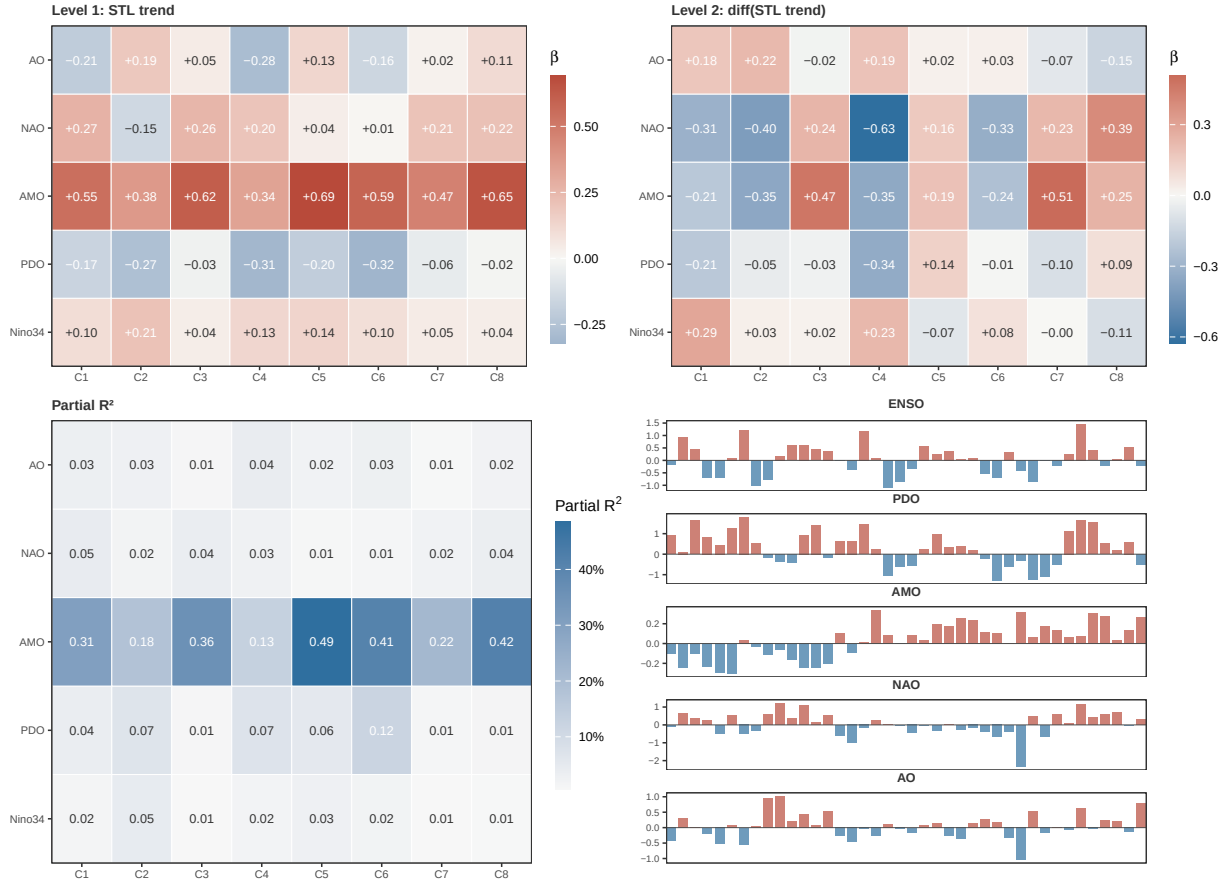


Figure 1: Teleconnection attribution by cluster: standardised coefficients, partial R^2 , and annual teleconnection index histories.

1.3 Level 2: Acceleration attribution

i Finding 16

The five indices explain less variance in the first-differenced STL trend (mean $R^2 = 0.26$) than in the trend level (mean $R^2 = 0.43$). This suggests that teleconnection indices are better at explaining long-term trend levels than year-to-year changes in trend slope.

i Finding 17

The standardised coefficients for the first-differenced model are uniformly small ($|\beta| < 0.1$ at the global level). This indicates that no single teleconnection index strongly drives the acceleration or deceleration of lake warming at the global scale.

1.4 Interpretation

The strong AMO signal across all clusters is consistent with the known role of Atlantic multidecadal variability in driving Northern Hemisphere temperature trends. The weak ENSO signal is expected: ENSO is an interannual phenomenon with limited persistence on the 40-year timescale of the STL trend.

The cluster-level variation in teleconnection response confirms that the eight temporal response types capture genuinely different climate sensitivities, not just statistical noise. Lakes in C5

(warming Europe) are most strongly influenced by AMO, while lakes in C4 (decelerating North America) are most strongly influenced by PDO.